



DNS 51 MSFM S2.5T Installation and Maintenance

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Chapter 1. Initial Setup

1. Overview

The Detectronic MSFM System is an ATEX and IECEx certified Hazardous Area Open-Channel Flow Monitor and data-logger with integral GPRS remote communication function and rechargeable external ATEX Li-Ion battery pack. It is an Area x Velocity Flow Meter with multi-sensor measuring elements that has a variety of access routes to its Data, including Web-based solutions.

The system is available in the following configurations dependent upon the application in which it is to be used:

MSFM Lite S2.5T: Ultrasonic Level/Pressure Transducer Interface

MSFM S2.5T: Ultrasonic Level/Pressure Transducer Interface, Temperature, Flow, Depth and Velocity

The MSFM S2.5T offers dual technology Pressure Depth + Ultrasonic Level measurement together with Ultrasonic Doppler Velocity measurement. When both the Depth + Level sensors are active, their data is recorded independently on discrete logger channels, giving dual sensor redundancy with its attendant measurement confidence increase.

The MSFM S2.5T contains an additional microcontroller (the DI-2.5) for advanced velocity, depth, temperature and flow computations. If required, this is accessible via the FBU, DI-2.5 cable adapter and the DI-2.5 configurator software.

Cello is the generic name for the GSM/GPRS data-logging and communication features integrated into MSFM S2.5T system. It may be used in some of the PMAC Lite program menus.

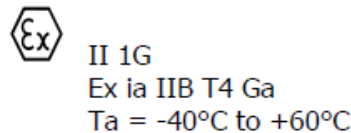
Each MSFM S2.5T supplied by Detectronic is issued with a unique 4 or 5 digit PMAC ID. This is normally already entered into the notepad of the logger and is the only ID to be used with the device whenever the data is being viewed on the various Detectronic web-hosting platform (DetecData-en, DetecData-de, DetecData Pro 2 and DetecAnalytics).

1.1 Safety Procedures

The following instructions apply to equipment covered by SIRA 19ATEX2255X.

Where applicable the information contained herewith shall be included, in full, in relevant certificate related Detectronic documentation.

Equipment marking:



The Multi-Sensor Flow Monitor S2.5T is used in flow monitoring applications in 'dirty water' environments such as sewerage. It combines a data logger and a mobile 'phone modem module and provides for external connections for several suitably-approved intrinsically safe sensors/transducers. It features an internal antenna as well as the provision for the connection of an external antenna. It also features a SIM card. It comprises a plastic enclosure housing two printed circuit boards, a component-approved (Sira 18ATEX2066U) intrinsically safe interface board and a partially encapsulated processor board.

MSFM S2.5T is powered by an internal user-replaceable non-rechargeable intrinsically safe Technolog '9V3000' battery pack (dual 3.9V lithium cells in parallel, component-approved - Sira 08ATEX2238U).

The entity parameters are as follows:

	Ui (V)	Ii (A)	Pi (W)	Ci (μF)	Li (mH)	Uo (V)	Io (mA)	Po (mW)	Co (μF)	Lo (mH)
CSO (JP1-CSO/ENC)	0	-	-	0	0	7.14	99	175	0.6	1
CSO (JP8-DIGI I/O)	5.88	-	-	0	0	5.88	5.7	8.4	0.6	1
Comms	5.88	-	-	0	0	5.88	26	38	0.6	1
Pulse/PC (Port J7 - Pins 1&2)	0	-	-	-	0.9	8.61	36	76	54	108
Pulse/PC (Volt free port - J7 pin 4)	5.5	-	-	-	-	8.61	1	2	54	142221
Vel (Depth)	0	-	-	-	-	8.61	270	799	50	1
Press/Lev	0	-	-	-	-	8.61	270	799	54	0.9
Ext. Power	12.6	2.708	6.142	0	0.019	0	-	-	-	-

Essential characteristics of tools that may be fitted to the equipment: During the installation, and any subsequent maintenance, the equipment may require the use of standard workshop type tools.

Instructions for safety:

Putting into service: The equipment shall only be put into service by suitably trained personnel in accordance with the applicable code of practice.

Use: When using the equipment the "Conditions of certification / special conditions for safe use" shall be adhered to.

Assembling and dismantling: The equipment does not require assembly or dismantling.

Maintenance (servicing and emergency repair):

When maintaining the equipment the "Conditions of certification / special conditions for safe use" shall be adhered to

Subject to the applicable code of practice, the enclosure may be temporarily opened and the internal battery pack replaced whilst the equipment is 'live' including when it is located in the hazardous area. Only a '9E3000' or a '9V3000' battery pack, manufactured by Technolog, is permitted as a replacement. These battery packs are intrinsically safe and may be replaced by the user in the hazardous area whilst the equipment is live

The equipment contains no customer replaceable parts and is not intended to be repaired by the user. Repair of this equipment is to be carried out by the manufacturer, or their approved agents, in accordance with the applicable code of practice

Regular periodic inspection of the equipment should be performed by suitably trained personnel in accordance with the applicable code of practice, for instance IEC / EN 60079-17, to ensure it is maintained in a satisfactory condition.

Installation:

When installing the equipment the "Conditions of certification / special conditions for safe use" shall be adhered to

The equipment is to be installed by suitably trained personnel in accordance with the applicable code of practice

The equipment may be installed in a Zone 1 or Zone 2 hazardous area with flammable gases and vapours with apparatus groups IIB and IIA and with temperature classes T1 or T2 or T3 or T4. The equipment is certified for use in atmospheric conditions in the range 80 to 100kPa (or equivalent), with up to 21% oxygen, in ambient temperatures in the range of -40°C to +60°C (or equivalent) and should not be used outside this range

If the equipment is likely to come into contact with aggressive substances, e.g. acidic liquids or gases that may attack metals or solvents that may affect polymeric materials, then it is the responsibility of the user to take suitable precautions that prevent it from being adversely affected thus ensuring that the type of protection is not compromised

Adjustment: With regard to safety, no user adjustment is required.

Danger areas in front of pressure-relief devices: There are no pressure relief areas.

Training instructions: With regard to safety, no training instructions are required.

Specific Conditions of use:

The Multi-Sensor Flow Monitor S2.5T has provision for the installer to connect an external antenna, which is not covered by the certificate. The circuit ground is connected to the exposed metal shell of this antenna connector, so the equipment does not meet the 500 Vac isolation requirements between circuit and enclosure. This shall be considered during installation.

The circuit ground is deliberately connected to the connector shells of the 'VEL' and 'PRESS/LEV' ports, so the equipment does not meet the requirements of the 500 V dielectric strength test in accordance with EN 60079-11. This shall be taken into consideration during installation.

Only a 9V3000 internal battery pack, manufactured by Technolog, is permitted as a replacement. These battery packs are intrinsically safe and may be replaced by the user in the hazardous area whilst the equipment is live.

Under certain extreme circumstances, the exposed plastic parts of the enclosure and the metallic connectors may store an ignition-capable level of electrostatic charge. Therefore, the user/installer shall implement precautions to prevent the build-up of electrostatic charge, e.g. locate the equipment where a charge-generating mechanism (such as wind-blown dust) is unlikely to be present and clean with a damp cloth. The metallic connectors have a capacitance of up to 26pF with respect to an earthed conductor.

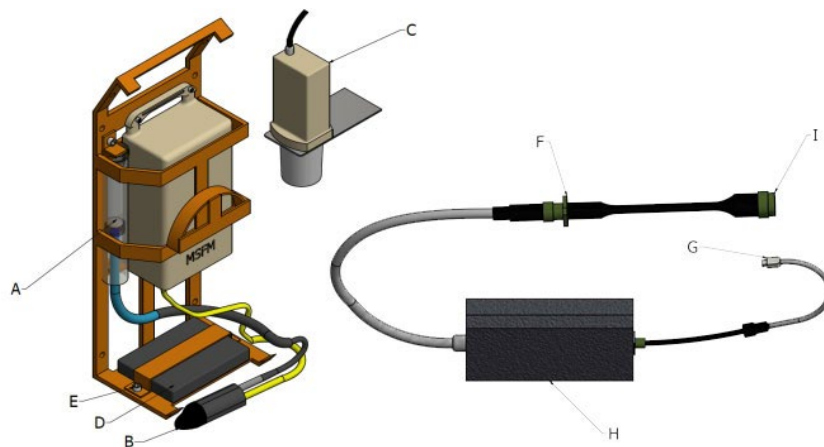
Some of the external metallic parts may be manufactured from light metal. In rare cases, ignition sources due to impact and friction sparks could occur. This shall be considered during installation, particularly if the equipment is installed in a zone 1 location.

The 'EXT PWR' port shall be connected to a resistive supply with a source resistance of 4.431 Ω minimum.

1.2 Components

Product dimensions: L 53 × W 21.5 × D 15 cm

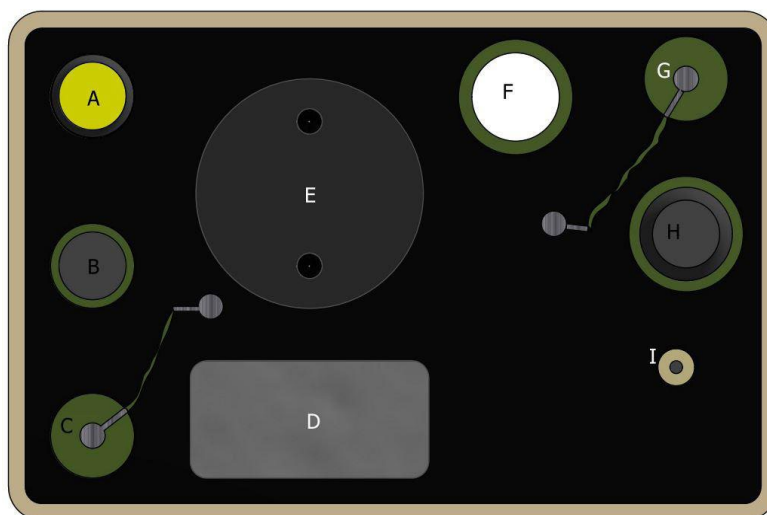
The following is a diagram of the MSFM S2.5T components:



Label	Component
A	Breather Assembly
B	Area Velocity Sensor
C	Ultrasonic CSO Monitor
D	ATEX 9W3000 or 9W3200 Lithium-Ion Rechargeable External Battery Pack
E	Allen Key Head screw
F	To MSFM 3 pin Comms
G	Local Comms USB – PC to Logger
H	FBU Barrier Module
I	DI-2.5 Adapter Cable (P123-PN018)

1.3 Connector Layout

The following is a diagram takes a closer look at the MSFM S2.5T connector layout:



Label	Component
A	Velocity Sensor Connection
B	Depth Sensor Connection
C	Connector for Ultrasonic Level Probe or Additional input and output adapter
D	Sealed SIM location
E	Internal battery (9V3000) access
F	DI-2.5 local comms connector & pulse output
G	logger local comms connector
H	External battery (9W6200) connector
I	External Antenna

2. Computer Setup

2.1 Equipment

To prepare the unit for installation the following is needed:

- A computer running Microsoft Windows 7 or newer versions
- A PC/Laptop with PMAC Lite software. This software can be obtained from a Detectronic representative or the support portal on the Detectronic website
- A USB port or 9 pin serial port
- An MSFM Fast Barrier Unit (FBU)
- DI-2.5 FBU adapter cable
- A 5 mm Allen Key is required for the removal /replacement of the ATEX Battery Pack and the Breather Assembly

2.2 PC Setup

1. Start the computer running the PMAC Lite program
2. Connect the MSFM local communications connector to the Laptop/PC serial input or USB input using a PC to the Logger and FBU communications cable. The Laptop/PC must **not** be taken into the hazardous area.

2.3 Navigation Within PMAC Lite

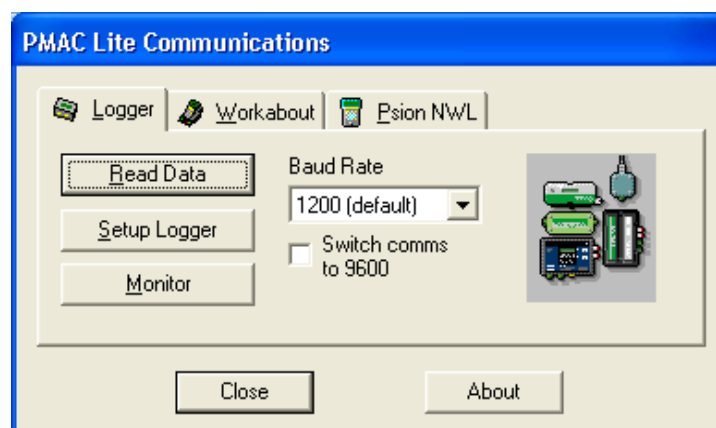
PMAC Lite operates as a floating menu bar that sits on the laptop or computer desktop and allows communication between the computer and the MSFM S2.5T.

The Cello Setup menu allows access to communications and setup for the logger.

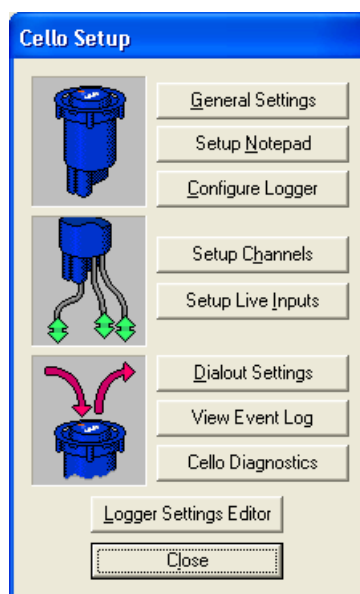
1. Once PMAC Lite is running, access the Cello Setup menu by clicking on the Communications icon  as shown below:



The following screen will appear:



2. Click "Setup Logger" and the Cello Setup menu will appear as follows:



Within this manual, instructions to navigate to this menu (i.e. following steps 1 and step 2) will appear as:

Communications → Setup Logger

2.4 Remote Communications Setup

Before taking the MSFM S2.5T to the site, it is recommended that Remote Communication features are set up and tested and that Automatic Data Send is enabled and available on the chosen delivery platform.

To check and test Remote Communications:

1. Login on DetecData-en, DetecData-de, or DetecData Pro2

A login name and password for each MSFM or suite of MSFMs is supplied. If this is not able to be found then contact Detectronic with the serial number of the MSFM S2.5T supplied and the order details. Detectronic will arrange to have the login setup. Please allow up to two working days for the login to be activated for website viewing.

2. One logged in, connect the MSFM S2.5T to a Laptop with PMAC Lite using the FBU interface and PC to Logger local serial communications cable
3. Run PMAC Lite on the Laptop/PC and navigate to:

Communications → Setup Logger

The following screen will appear:



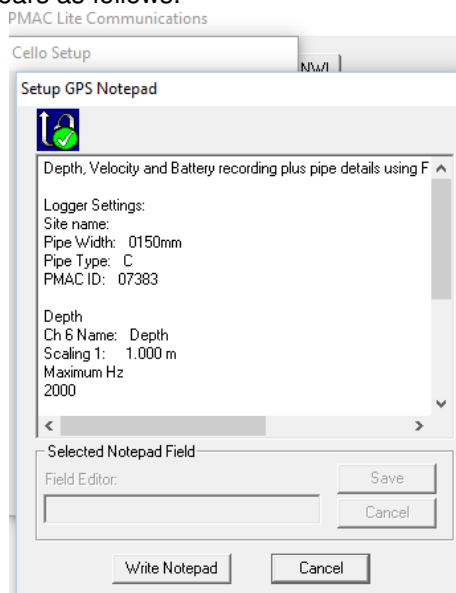
4. Tick the box marked “Switch comms to 9600”. This will ensure all local communication is done at fast speed where possible

2.5 Setup Notepad

Setup Notepad is used to program the unit with site details. It is accessed by navigating to:

Communications → Setup Logger → Setup Notepad

The following screen appears as follows:



In order to edit the fields, undertake the following steps:

1. Click on the desired field to be edited
2. Type details in the field editor area using the keypad
3. Click “Save” after editing each field
4. “Write Notepad” **must** be clicked once all the required fields have been amended to save to the logger
5. After finishing editing the details, the Notepad should always be returned to in order to check it is set correctly before continuing

Site Name

It is advisable to enter the Site Name at this stage for ease of identification. This Site Name is displayed on the website and should be unique amongst the sites belonging to the user and should not contain any special characters such as "?", "!", "%", etc.

Pipe Width

For a circular pipe, enter the pipe diameter. For a rectangular pipe, enter the width of the channel. All measurements entered will be recorded as millimetres. Leading zeros **must** be entered. For example, "0150 mm" must be entered for a 150 mm width pipe.

If the pipe is neither circular nor rectangular, contact Detectronic.

Pipe Shape

Enter "C" in the Pipe Type for a circular pipe. Enter "R" for rectangular shape channels. For other pipe shapes, contact Detectronic.

Entering PMAC ID

The notepad PMAC ID number is unique to each MSFM S2.5T and should not be changed at any time. This number will already be entered using the PMAC ID provided to the user by Detectronic.

Velocity, Pressure Depth, Temperature, and Battery Voltages Measurements

Velocity, Depth, Temperature, and Battery Voltages measurements are pre-set up and pre-calibrated when the unit is supplied. No setup should be necessary.

Ultrasonic Level Measurement Output Range

If the user has an Ultrasonic Sensor, the Ultrasonic Level Measurement Output Range allows the user to set the required parameters for it.

It can be setup before being installed, provided that the exact distance between the face of the Probe and the Empty Distance in the channel or pipe is known. See Appendix A for instructions on positioning the probe and calculating the Output Range to enter in the Notepad.

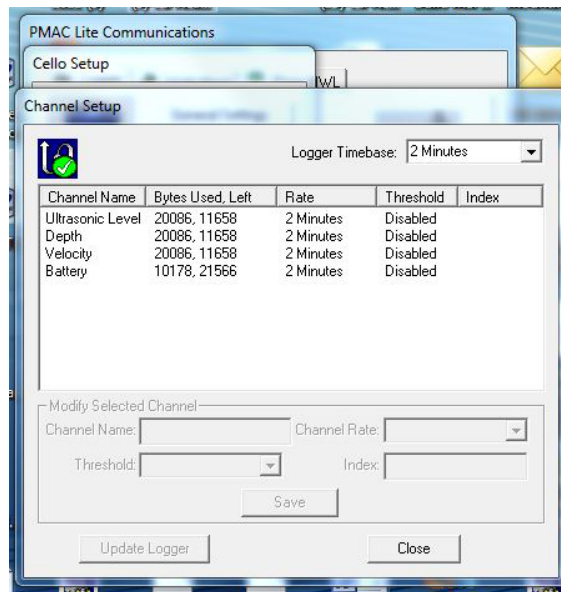
2.6 Setup Channels

The Channels within Channel Setup refers to the data channels that readings are taken from. The Logging Interval is the regularity that the logger takes a reading.

1. To set the Logging Interval, navigate to:

Communications → Setup Logger → Setup Channels

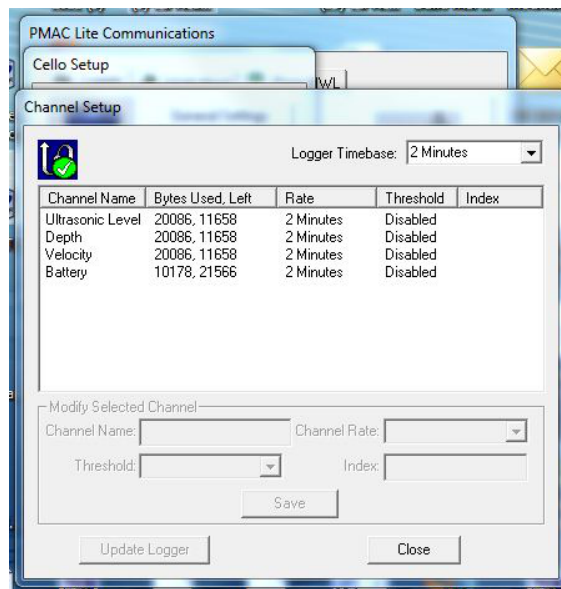
The Channel Setup appears as follows:



2. Use the drop-down menu of the Logger Timebase to select the Data Logging Rate. The individual channels will adjust in line with this timebase. Logging rates can be anything between 1 minute and 1 hour.

If channels are required at different logging rates then change each channel individually by clicking on the channel and altering the Channel Rate in the lower “Modify Selected Channel” section. The Depth and Velocity Channels **must** be on the same logging rate for the flow information to be calculated correctly.

An example of individual channels being modified is displayed below:



3. Press “Update Logger” to save the changes to the logger

2.7 Dialout Settings

The Dialout Settings menu and Advanced Settings are used to set all the Remote Communications options including dial-out frequencies and alarm levels.

To set up the Dialout Settings for Remote Communications navigate to:

Communications → Setup Logger → Dialout Settings

The Dialout Settings appears as follows:

Logger Dial-Out and Alarms Editor

Regular Wake-Up Options

☒ Enable regular wake-up

Wake-Up Hour: 05 Min: 01

Alarm Configuration

Channel Name	HIGH Alarm	LOW Alarm	Dialout Options
Depth	Disabled	Disabled	SMS
Velocity	Disabled	Disabled	SMS
Battery	Disabled	Disabled	SMS

Selected Channel Alarm Settings

High Alarm: [dropdown] [dropdown]

Low Alarm: [dropdown] [dropdown]

Alarm Window Times

Alarm Window Start: 00 End: 00

Dialout Numbers

PMAC Dialout 1: gprs.utidat.net

PMAC Dialout 2: [blank]

PMAC Dialout 3: 1801

SMS Data: [blank]

Modulo Number: [blank]

Own Number: +1555171326347

Buttons: Advanced Settings, Update Logger, Cancel, Update List, Cancel List Edits

4. Tick “Enable regular wake-up” to activate the daily data send. If this is not ticked, the MSFM S2.5T will have no scheduled wake-up time and will only log data rather than logging and sending data
5. Enter the fields for the top right section of the Logger Dial-Out and Alarms Editor as shown below:

PMAC Dialout 1: “gprs.utidat.net” unless data is going direct to the user’s own server

PMAC Dialout 2: (leave blank)

PMAC Dialout 3: “1801”

SMS Data: (leave blank)

Modulo Number: (leave blank)

Own Number: Contains the unique phone number for the enclosed SIM card and should **not** be changed

6. Click “Update Logger”
7. Click “Advanced Settings” to access the Advanced Channel Alarm Settings. The screen appears as follows:

Advanced Channel Alarm Settings

☐ Send Alarm using a data call

☒ Send Alarm using an SMS

☐ Send Modulo profile alarms

☐ Just alarm - send no data

☐ Also send latest data for this channel

☒ Also send latest data for ALL channels

☒ Repeat sending latest data:

Number of Repeats: 1

Interval between repeats (mins): 60

Close

8. Keep the default option “Send Alarm using an SMS” selected. This option will cause the data to be sent via the standard alarm transmission method. Enter the rest of the required settings. The default Number of Repeats and Interval between repeats sends the latest data for all channels to repeat once every 60 minutes

“Send Modulo profile alarms” is not an option applicable to equipment supplied by Detectronic, though may be used in conjunction with equipment from other suppliers.

9. Click “Close”
10. Click “Update Logger” to update the settings and the Setup Menu will be displayed, confirming that the settings have been saved to the Logger
11. Navigate to:

Communications → Setup Logger → General Settings

The screen should appear as follows:

12. Press “Start Logging” to start the logger recording
- Once logging has begun, the MSFM S2.5T will send a copy of its settings file to the server. This can be verified by going online to the appropriate website. At this stage only the Site Name and logging parameters will be registered.

2.8 Cello Diagnostics

Cello Diagnostics allows the user to access diagnostic information relating to the logger. It can be used to check that the MSFM S2.5T is sending data. The Send Latest Data function can be done before or after installation as necessary. Leave the MSFM S2.5T in Logging Mode for a few logged intervals data and undertake the following steps:

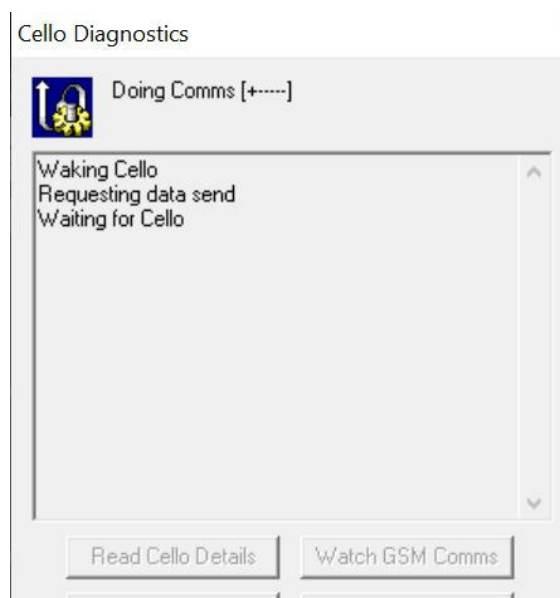
1. Navigate to:

Communications → General Settings → Cello Diagnostics

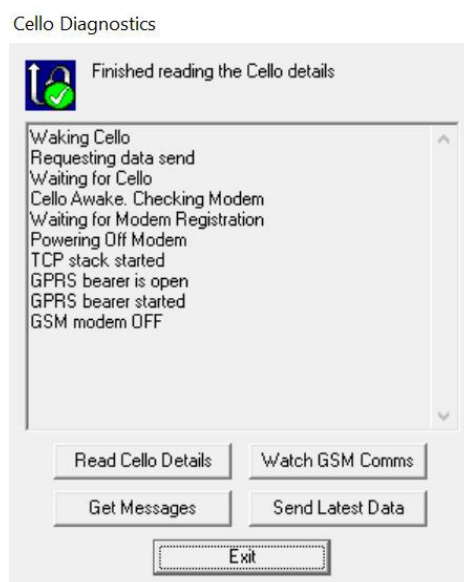
The screen will appear as follows:



2. Press "Send Latest Data". The screen should appear as follows:



3. Wait until the program confirms finishing, as shown in the example below:



4. To confirm the data has been sent, check on the relevant DetecData website for the data upload. This may take up to 10 minutes depending on Network Traffic. The address of this site will vary depending on the user's region and will be supplied with the user's package. If sensors are out of water, zero readings will be logged and sent to the DetecData website
5. Wait for "Modem Off" to be shown on the Cello Diagnostics screen once the modem within the logger has finished sending. Once this is visible and the data has been confirmed on the website, the MSFM S2.5T is ready to be prepared for site installation if it has not already been installed

3. Preparing For Site Installation

MSFM S2.5Ts are pre-configured to be specific to the site prior to the user receiving them and the units measured are tailored to the user's requirements. Users with Ultrasonic Level Sensors may wish to configure the logger to include the Level Channel.

MSFM S2.5Ts can be pre-programmed in offices or workshops to verify the sensors and establish GPRS comms before going to the site. Installation can take place without the need for local communication though, if possible, local checking **must** be undertaken. This is most often undertaken by doing a local comms sanity check on the velocity and a local comms depth check using a rule to get the exact measurement.

3.1 Input Configuration

1. Connect the MSFM S2.5T to a PC/Laptop running PMAC Lite via the Local USB Comms
2. Run the PMAC software
3. Navigate to:

Communications → Setup Logger → Setup Live Inputs

If this is done with the MSFM S2.5T in standby mode before Data Logging is started then the procedure outlined below can be followed. If the MSFM S2.5T is already logging then when each parameter is selected, the last logged value will be displayed **instead** of a live current value. The last-logged value will be updated at the data-logging interval, for example, when Logging every 5 minutes, the "Read Input" displayed value will only update when the next 5 minute measured value is logged. This applies to all channels when reading input.

In the Input Configuration menu, the first channel available to view will be the Ultrasonic (CSO) Channel or the Pressure Transducer Interface if this is configured. See Appendix A for the Ultrasonic Level Sensor setup, and Appendix B for the Pressure Transducer Interface setup

4. Navigate between the different channels by clicking on "Next Channel" and "Prev. Channel". The different channels can include Internal Battery, Water Temperature, Flow, Pressure Depth, Velocity, and External Battery. Each channel is elaborated after these steps and appears similar to as follows:



5. Once finished, click "Close" in the Input Configuration Window and prepare for site installation (see section 3 of this chapter)
6. Following Installation, it is recommended that a calibration check of the readings is performed by using a certified reference Velocity Meter (Velocity Profile Check) and

Ruler (Dip Check) to check the calibration of the installed Sensors. Once calibration checks have been performed, navigate to:

Communications → Setup Logger → General Settings

The screen will appear similar to as follows:

Logger Information

Logger Type: Cello 3.77 (standby) Start Logger

Memory Size: 128K

Logger Time

Logger Time: 11:46:00 (20/07/2009) Sync to PC Time

PC Time: 11:46:04 (20/07/2009)

Editable Settings:

GPS Protocol ID: 0 Update Logger

Timebase: 2 Minutes

Time Resolution: 10 Seconds

Store Mode: Rotating Store

Close

- Click "Start Logger" to begin recording

Internal Battery

Navigate to the Internal Battery channel using the Input Configuration as described in Step 4 on p17. Click "Read Input". View the Current Value field to check the Internal Battery voltage. For new units this should be 3.6V or above.

Water Temperature

Navigate to the Water Temperature channel (the fourth channel) using the Input Configuration as described in Step 4 on p17. Click "Read Input". View the Current Value field to check the water temperature. Users should note that the temperature measurement may take some time to settle after submersion.

Flow

Navigate to the Flow channel (the fifth channel) using the Input Configuration as described in Step 4 on p17. Click "Read Input". View the Current Value field to check the flow. Users should note that the flow is calculated by the DI-2.5; if the flow is incorrect but depth and velocity are correct, the channel type or dimensions may need checking. (Flow may be calculated on the website instead of with the logger and the flow channel may not be visible locally.)

Pressure Depth

Navigate to the Pressure Depth channel using the Input Configuration as described in Step 4 on p17. The Depth reading should either be zero or very close to it. Values of 0.002 to

0.003m are acceptable while in air.

Velocity

Navigate to the Velocity channel using the Input Configuration as described in Step 4 on p17. The reading should be zero with sensor out of water.

External Battery

Navigate to the Battery channel using the Input Configuration as described in Step 4 on p17. View the Current Value field to check the battery voltage is above 12.2V for new installs. This is to ensure the external battery is sufficiently charged to power the sensors.

3.2 Local Data Download

After stopping the logger, the data stored in the logger will be lost when logging is restarted.

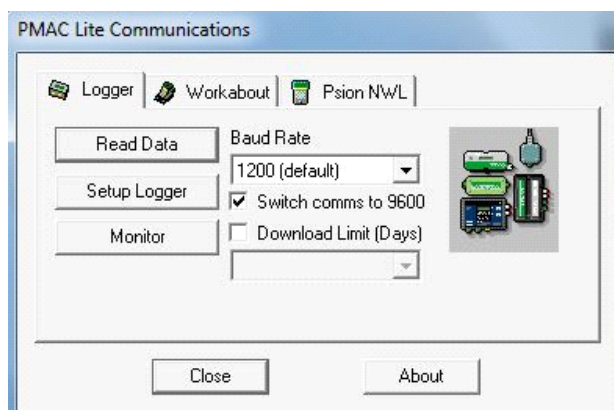
When local data download is required (or before the Data Logger is stopped and restarted), it is important to either download and save the logged data or to send it to DetecData using the Send Latest Data function in the Diagnostics menu.

The Local Download procedure allows local data retrieval on site and is achieved by undertaking the following steps:

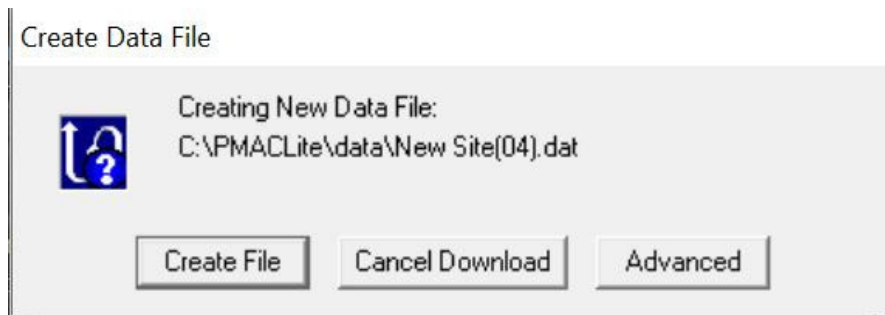
1. Connect the laptop with the PMAC Lite program to the logger using the MSFM S2.5T communication cable
2. Within PMAC Lite, Navigate to:

Communications → Setup Logger

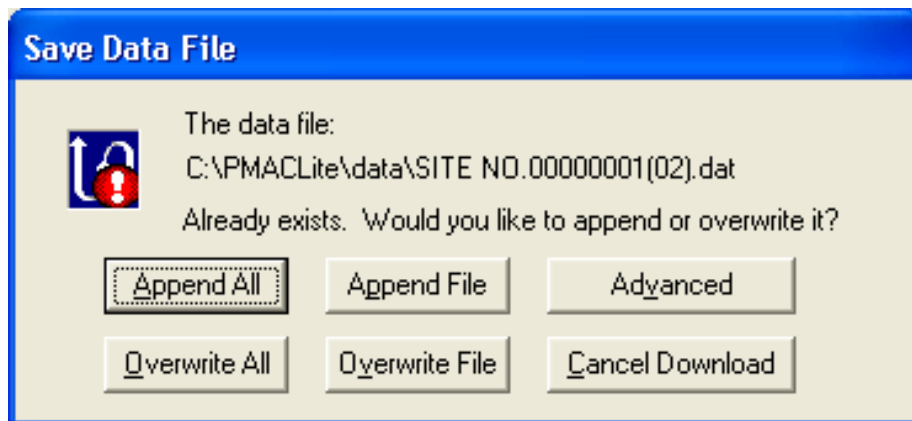
The following screen will appear:



3. Click "Read Data" If PMAC Lite has **not** downloaded data from the site before then proceed to Step 4. If PMAC Lite **has** downloaded data from the site before then proceed to Step 5
4. The screen should appear as follows:



Click "Create File" to complete the download. For additional Advanced options, see Step 6 5. The screen should appear as follows:



Select the appropriate option from the following:

Append All: Add all downloaded data to all existing files for all channels

Append File: Add downloaded data to files for this channel only

Overwrite All: Overwrite all existing files for all channels

Overwrite File: Overwrite existing files for this channel only

Advanced: (see Step 6)

6. Clicking "Advanced" allows users to alter preference options for data storage. Save any altered settings by clicking "OK". The Advanced screen appears as follows:

 A dialog box titled "Enter Site Details" with a blue header bar. It contains three sections: "Site Name", "Channel Names", and "File Names".

 - "Site Name": "Enter or browse for the Site Name that you want the logger's data to refer to:" followed by a text box containing "SITE NO.00000001" and a "Browse" button.

 - "Channel Names": Two radio buttons. The first is selected: "Use default/logger's channel names: If the logger contains channel names, they will be used. If the logger doesn't contain any channel names, then the channel names will default to 'Channel 1', 'Channel 2', etc." The second is "Use individual channel names:" followed by a "Set Names" button.

 - "File Names": "Either enter manually or browse for the file name that you want to use for the first logger data file:" followed by a text box containing "SITE NO.00000001" and a "Browse" button. Below this is "Example Filename:" followed by a text box containing "SITE NO.00000001(02).dat".

 At the bottom, there is a "PLEASE NOTE:" section: "A separate file will be created for each channel of data in the logger. The files will all be called '<filename>[XX].dat' where <filename> is the name you specify and XX is the channel number."

 At the very bottom are three buttons: "OK", "Cancel", and "Help".

Site Installation and Maintenance

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Chapter 2. Site Installation and Maintenance

1. Safety Procedures

It is assumed that all the necessary system design, specification and engineering factors have been taken into account by suitably competent personnel before installation begins. Detectronic Ltd is dedicated to the design and manufacture of environmental monitoring equipment. Because of our policy of continuous product development, products may differ in detail from the information supplied in this manual. Different versions and editions of manuals can be found in the login area of the Detectronic website or through contacting a Detectronic representative. Detectronic Ltd has taken care in preparing these instructions but can accept no responsibility for any loss or damage caused either directly or indirectly by the use of software and/or equipment it supplies other than in accordance with this manual.

Ensure the following before installation:

- Installation is completed in accordance with the Sira 19ATEX2255X certificate
- The safety documentation confirms the proposed system is fully certified
- The components and equipment are the correct types as specified in the safety documentation
- All hazardous area cables are secured and segregated from all other cables
- All identification labels and markings have been inspected and checked to ensure the equipment is suitable for the installation location
- A final check of all the above points is done by another person as well as the person who initially ensured all of the above checks

2. Installation

Select the location for the Area Velocity measurement sensors, avoiding mounting near bends or additional inlets. Always make sure the sensor is placed in the bottom of the pipe, or if unable to clear any silt in the pipe then sensor can be off-set. This will need additional setup to compensate for the sensor depth off-set. Please contact Detectronic for details.

An optional 3rd sensor for both circular and rectangular channels allows the connection of an Ultrasonic Level CSO Probe (detailed in Appendix A) or Pressure Transducer Interface (detailed in Appendix B) for additional recorded information.

2.1 Tools and Equipment

The following tools and consumables are recommended for installation and maintenance:

Small screwdriver for 3mm slot head screws (also cross head screw driver may be required)
Medium screwdriver for 4 mm slot head screws (also cross head screwdriver may be necessary)

Small instrument screwdriver for depth span adjustment if necessary

Plastic Cable ties approx:

100 mm length by 2.5 mm wide - qty: 20

300 mm length by 4.6 mm wide - qty: 5

Allen Key 5mm ("T" bar type)

Snips (wire cutters)

Mole grips

Ratchet spanner with 17mm socket

Roll of strong adhesive tape 50mm wide approx.

Detectronic Protective Wax spray for connectors.

Suitable Digital Camera, for site records.

Three Digit Cut Cat 5 gloves (Cat 5 gloves must be worn when handling the metal ring/scissor expander)

2.2 Installation in a Circular Channel or Pipe

1. Using the screws provided, secure the sensor head to the sensor mounting plate as shown in Figure 1:



Figure 1: Underside of AV Sensor with a Plate

A: Ring mounting nut and screw holes M4
B: sensor fixing M3 × 10mm threaded screw holes
C: Pressure Port Hole (**do not block**)

2. Secure the sensor mounting plate to the stainless steel fixing ring with the screws provided as shown in Figure 2:

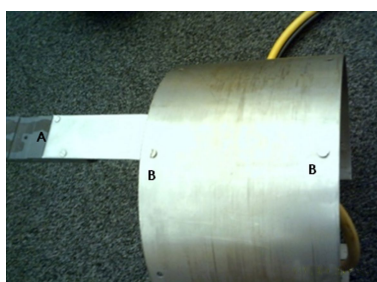


Figure 2: Sensor Mounting Plate attached to the bottom of the Velocity Sensor

A: Port pressure hole (**do not block** otherwise it interferes with pressure readings)
B: Two M3 by 10mm stainless steel screws

3. Feed the small cable ties through the small holes in the fixing ring to tie the transducer cables to the trailing edge of the fixing ring, as shown in Figure 3:

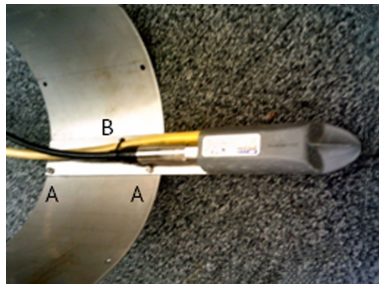


Figure 3: Top View of Sensor

A: the Sensor Mounting Plate attached to the ring with two M4 nuts and screws

B: small cable ties

4. Bend the cable to the circumference of the fixing ring as shown in Figure 4. Do not bend the cable sharply. Continue to feed the small cable ties through the holes to fit the cable to the trailing edge of the fixing ring

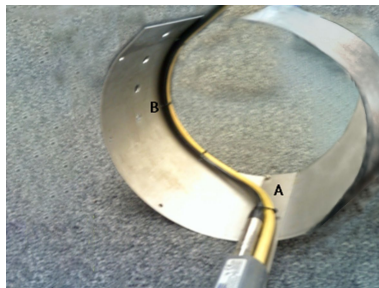


Figure 4: Bending the Cable to the Fixing Ring

A: the cable is not bent sharply

B: the cable ties fit to the edge of the fixing ring

5. Ensuring the pressure port hole remains unblocked, wrap the cables and fixing ring with strong and waterproof self-adhesive tape. This will present a smooth surface to the flow of liquid behind the transducer head and minimise the risk of ragging
6. If a scissor expander is required (usually for larger rings over a 300mm diameter), secure it to the ends of the fixing ring with the stainless steel split pins provided as shown in Figure 5. Figure 5 shows the AV sensor position at the bottom of the ring below the expander. Split pins are used to hold the scissor expander to be fitted to the ring ready for insertion into the open end of the pipe, which is usually done after the sensor and cable is secured to the ring

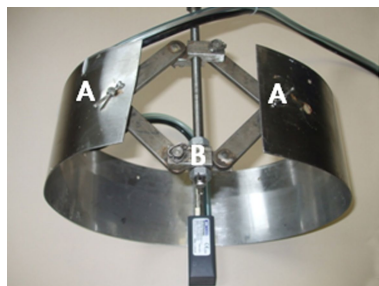


Figure 5: Stainless Steel ring with the Scissor Expander fitted

A: Split Pins

B: A Socket Spanner with extension is used to extend the expander to fix the ring firmly when it is in the pipe

7. Insert the ring into the pipe so it is secure and has the transducer head facing into the flow. Ensure the transducer head is flat on the bottom of the pipe and is perpendicular to the flow. Depending on the pipe size, the fixing ring may need to be compressed manually or expanded with a scissor expander. See an example of an installed Sensor Head in Figure 6:



Figure 6: Cables Without Duct Tape in a Circular Pipe

2.3 Installation in a Rectangular Channel

1. Use an “L” or “U” shaped bracket or a flat bracket designed to be fixed to the base or invert of the channel. The optimum installed position for Velocity is typically a third of the channel width from either of the sides
2. Fit the transducer to the bracket using the countersunk screws provided
3. Using small cable ties, tie the transducer cables to the trailing edge of the bracket to minimise ragging
4. Wrap the cables and fixing ring with strong and waterproof self-adhesive tape. This will present a smooth surface to the flow behind the transducer head and to further minimise the risk of ragging



Figure 7 (left): Stainless Steel Rectangular Channel L-Shaped Mounting Bracket With Cable Wrapped with Tape



Figure 8 (right): Stainless Steel Rectangular Channel U-Shaped Mounting Bracket With Cable Wrapped with Tape

2.4 Fixing The MSFM Main Unit

1. Mount the MSFM as high as possible in the manhole chamber. This is to minimise the risk of submersion in surcharge conditions and to ensure optimum GSM/GPRS signal transmission and reception. The MSFM can be mounted using a fixing spike or fixed to suitable step irons

The following image is an example of the MSFM mounted using a fixing spike:



The following image is an example of the MSFM fixed to suitable step irons using large cable ties and the unit's frame:



2. Route the transducer cables neatly coiling any excess tidily but not too tightly, then tie using large reusable cable ties to either the MSFM frame (the lower half only) or to a convenient step-iron

If installing the MSFM at a site where surcharge is possible, it is particularly important to ensure that the MSFM remains vertical at all times in order for its high-specification breather-system to perform as designed.

3. If water level may exceed the top of the MSFM then ensure MSFM is secured to a mounting point to avoid it floating off

3. Maintenance

Breather Desiccant Tube replacement, sensor cleaning, ATEX Battery Pack replacement, and Battery re-charging can be carried out by the user. All of these should be scheduled regularly at intervals appropriate to the application. The regularity of replacement will depend upon factors such as channel conditions, the dampness of the atmosphere, and how often the sensor is powered up. The external battery voltage level is logged as a channel of data and visible on DetecData-en, DetecData-de or the DetecData Pro2 website. Detectronic normally schedules the battery to be swapped when the voltage drops below 11V.

The unit contains no other replaceable parts and should not be opened. All repairs and upgrades must be carried out by the manufacturer or approved channel partners. Regular inspection of the equipment should be performed by suitably trained personnel to ensure continuing safe and satisfactory operation and cleanliness.

3.1 Charging the 9W3000 or 9W3200 Battery Pack

The battery **must not** be charged in a hazardous area.

The battery can either be charged in the field by removing the battery-pack from the MSFM using a 5 mm Allen key and recharging using a fully-charged 9W3000 or 9W3200 pack, or it can be recharged in the workshop by following the instructions below.

1. In the workshop, disconnect the battery-pack from the MSFM as shown in the image below:



2. Connect the battery-pack to the charger connector by turning it clockwise until it locks into position as shown:



3. In a **safe area**, plug the charger into the mains supply and monitor the charger status by observing the LED colour. Orange and yellow indicates that the battery is being charged, while green indicates that the charge cycle is complete and that the battery is fully charged and ready for use



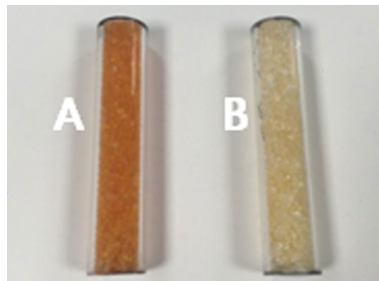
*Figure 8: The Battery Charger
With the Guide to LED Colour
Indicating the Battery Status*

3.2 Cleaning the MSFM

The MSFM and its Sensors should only be washed with warm (not HOT) soapy water using a soft brush or cloth - then gently rinsed with either clean water or a mild disinfectant solution. Never power wash the MSFM or any of its sensors.

3.3 Changing the Breather

If the desiccant crystals in the breather tube are showing signs of changing from orange to white (as shown in Figure 9) then the desiccant within the tube needs replacing if total moisture protection of the Pressure Depth Sensor is to be maintained.



*Figure 9: Desiccant Crystals
in the Breather Tube*

A: Orange crystals that don't need replacing
B: Crystals that have turned white that need replacing

In order to change the breather follow the instructions below:

Disconnect the transducer cables from the unit to enable access to the breather connection. Using a small cable tie, secure the blue-green tubing approximately halfway across its length to the Pressure Transducer cable as shown in Figure 10. This prevents stress to the crimped joint.



*Figure 10: Attaching the
Blue-Green Tubing to the
Pressure Transducer Cable*

A: Cable tie
B: Crimped joint

4. Carefully remove the breather pipe from the Breather. The best technique for this is for the user to hold the tube with their fingers and place their thumb against the clear

breather tube, then to gently and slowly pull the tube off the breather while using the thumb as a lever pushing against the breather, as shown in Figure 11:

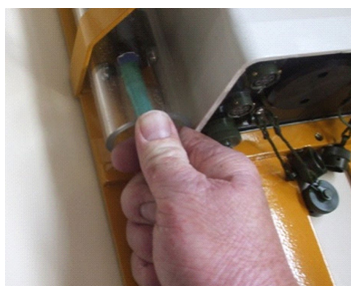


Figure 11: Removing the Breather Pipe From the Breather

Unscrew the retaining bracket and the unit frame mounting screws using a 5mm Allen key, and remove the breather

5. Reattach a new Breather to the cable; a small amount of grease may assist this
6. Refit the breather to the Q-Flow and re-tighten the breather retaining bracket
7. Remove the securing cable ties as appropriate

3.4 Replacing and Calibrating the Pressure Sensor

It is recommended that the Pressure Sensor is returned to Detectronic for calibration and replacement. Local calibration can only be done using the separate Detectronic configurator software by approved and trained personnel

3.5 Replacing and Calibrating the Velocity Sensor

The Velocity Sensor must **not** be replaced on site as it is tuned to the interface board within the logger. If the Velocity Sensor requires replacing, contact Detectronic.

A Velocity Sensor Calibration or check will require placing the sensor in an open-channel flow greater than 0.150m wide, with a controllable, known, and stable Velocity Range between 0.10 m/s and 1.50 m/s, and with suitable water depths greater than 0.10m

4. Temperature

The MSFM S2.5T will measure water temperature on logger Channel 4. The measured water temperature can also be used to internally to compensate for variations of velocity and pressure due to temperature variation.

5. Flow Calculations

It is possible to enable the MSFM S2.5 to calculate flow and/or provide a flow proportional pulsed output using the internal DI 2.5 board. The DI-2.5 is capable of calculating flow based on depth, velocity and channel geometry. The DI-2.5 must be setup independently of the logger using the DI-2.5 configurator software and be connected to a laptop or computer using an FBU via a DI-2.5 adapter cable.

The following channel geometries are possible:

- Rectangular (width must be entered)
- Circular (pipe diameter must be entered)
- Trapezoidal (base width, upper width, height to upper width required)

6. Pulse Output

A pulse output is available to enable flow proportional sampling. The output is connected via a pin on the DI-2.5 local communications connector.

If the MSFM is located in a hazardous area, then the pulse output must **only** be connected to suitably certified equipment.

The output is set using the DI-2.5 configurator program.

Appendix A: Ultrasonic CSO Level Sensor with MSFM S2.5T and MSFM Lite S2.5T Setup

The CSO Level Monitor is specifically designed to measure combined sewage overflows (CSOs) and it can also be used for remote monitoring of reservoir and tank levels, including those located in potentially explosive atmospheres.

If the CSO Level Sensor is to be installed then connect the CSO probe to its connector on the MSFM manifold, ensuring that the two redundant dust-caps are also connected together for storage.

i. Equipment

The following equipment is required:

- A laptop with PMAC Lite installed will be required on site
- Local comms. FBU ATEX barrier with CBL205 PC to logger comms cable and USB adapter
- A 0-3m length tape measure
- Two spanners for size M5 Nuts and Bolts (for stainless steel mounting plate adjustments) or adjustable spanners

ii. Empty Distance Measurement in a Manhole Shaft

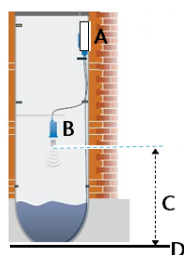


Figure 13: Measuring the Empty Distance in a Manhole Shaft

- A: MSFM or MSFM Lite
- B: CSO V2 Monitor
- C: Empty Distance
- D: Bottom of channel

1. Measure the Empty Distance. The Empty Distance is the distance from the sensor head down to the bottom of the dry channel

The Empty Distance in Figures 13-15 is 1000mm. This includes the 300mm Deadband. The Deadband is the minimum distance away from the sensor head required to obtain a sensible measurement as within the Deadband the signal does not propagate properly.

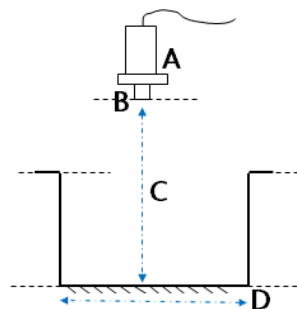


Figure 14: Probe Position in a Rectangular Sewer Pipe

- A: CSO Monitor V2 Probe
- B: Probe Face
- C: Empty Distance
- D: Channel Width

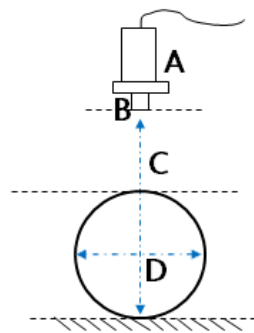


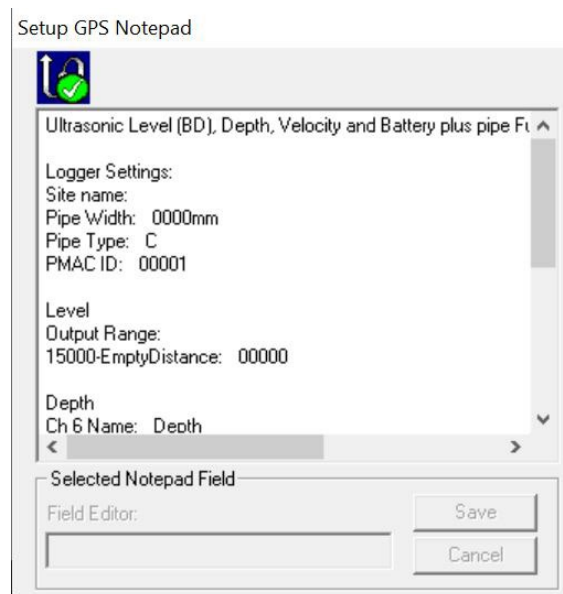
Figure 15: Probe Position in a Circular Sewer Pipe

A: CSO Monitor V2 Probe
 B: Probe Face
 C: Empty Distance
 D: Pipe Width

2. Within PMAC Lite, navigate to:

Communications → Setup Logger → Setup Notepad

The screen should appear similar to as follows:



If the Level section is not present then reconfiguration will be required so contact Detectronic for more details.

Do **not** alter the Output Range.

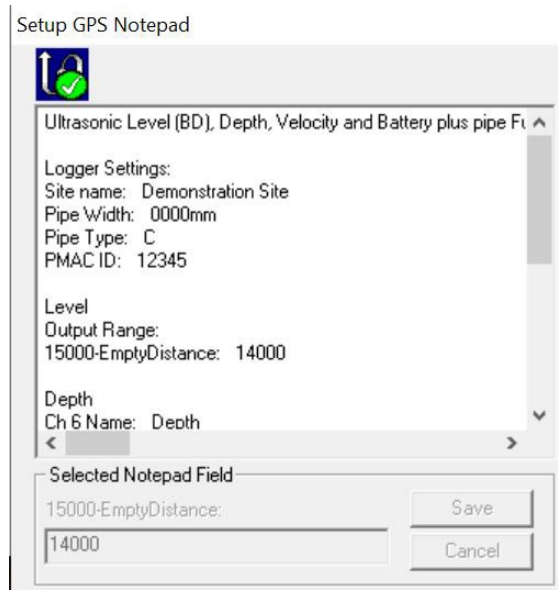
3. Calculate the Current Level. The calculation for the Current Level for any site will be:

15000 mm - the Empty Distance (mm)

Therefore in the example used, the calculation would be:

$$15000 \text{ mm} - 1000 \text{ mm} = 14000 \text{ mm}$$

4. Enter the Current Level into the "15000-EmptyDistance" field. In the example used, the number entered for the Current Level would be 14000, as shown:



5. Save by clicking "Write Notepad"

iii. Installation Notes

For channels with an open top (e.g./ manhole access), the probe must be placed in a location without any obstructions between the face of the CSO V2 Probe and the bottom of the channel.

The example in Figure 16 shows the basic CSO V2 setup for a water level measurement up to the top of the pipe (PIPE ID) and assumes the sensor has a clear view to the channel with no obstructions in the way. Obstructions between the sensor and channel would result in incorrect readings due to the ultrasonic signal bounce off.

To measure surcharge, place the face of the CSO probe 300 mm above the highest discharge level in the sewer, then measure the Empty Distance between the face and the empty position in the sewer pipe (i.e. the bottom of the pipe).

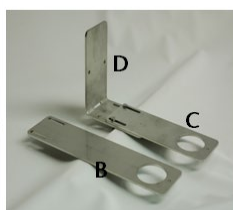


Figure 16: CSO Mounting Plate Options

- A: CSO mounting plate Short option
- B: CSO mounting plate Long option
- C: Short mounting plate
- D: 90° Stainless Steel wall mounting bracket option

If the channel or chamber is so shallow that the recorded level would be within the Deadband then a 45 degree reflecting plate may be used with sensor positioned horizontally, as shown:



Appendix B: Pressure Transducer Interface (PTI) with the MSFM S2.5T and MSFM Lite

The Pressure Transducer Interface is an additional depth sensor that allows the measurement of water depth in applications where traditional ultrasonic measurement is not appropriate such as for sites with turbulent or interrupted flow, wells and boreholes, CSOs with high spill points, weirs and flumes exposed to the elements, and large range sites.

1. If the MSFM setup using PMAC Lite has not been undertaken, follow through the steps from Chapter 1. The software version of PMAC Lite **must** be the 2014 version or a later release in order to setup the Pressure Transducer Interface
2. Run the PMAC Lite program
3. Navigate to:

Communications → Setup Logger → General Settings

The screen should appear as follows:

Logger Information

Logger Type: Cello 4.06 (standby) Start Logger

Memory Size: 128K

Logger Time

Logger Time: 13:48:16 (30/04/2020) Sync to PC Time

PC Time: 14:48:37 (30/04/2020)

Editable Settings:

GPS Protocol ID: 0 Update Logger

Timebase: 2 Minutes

Time Resolution: 10 Seconds

Store Mode: Rotating Store

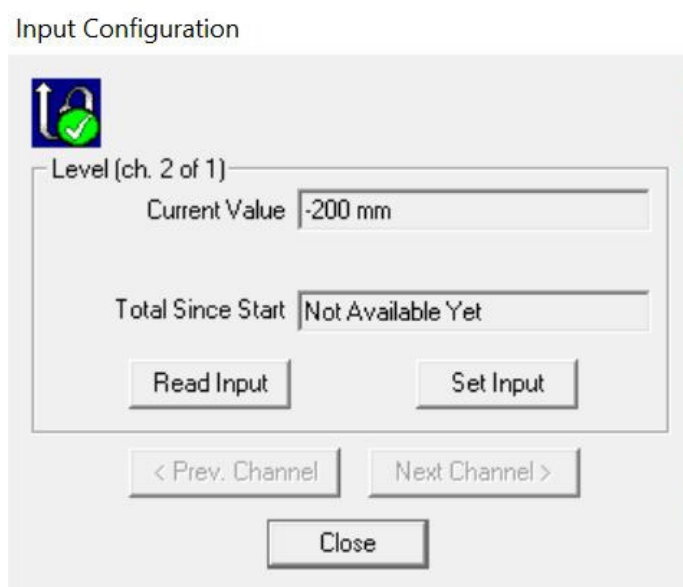
Close

4. Check the Logger Type of the MSFM is v4.01 or greater. If it is, then proceed to Step 5. If it isn't, then contact Detectronic for upgrade instructions
5. Navigate to:

Communications → Setup Logger → Setup Live Inputs

The screen should similar to as follows over the page:

Input Configuration



Level (ch. 2 of 1)

Current Value -200 mm

Total Since Start Not Available Yet

Read Input Set Input

< Prev. Channel Next Channel >

Close

6. Adjust the Offset to read the required depth by clicking on “Read Input” to get the Current Depth, then by clicking on “Set Input” and adjusting the Current Value. An example is shown below:

Set Input Values

Please enter the new input value in the space provided.
Hit cancel to exit without making changes.

Current Value -200 mm

Write Changes Cancel Zero Input

7. Click “Write Changes” to save the adjustments to the Offset. Once this done, adjusting the Dialout Settings, setting the clock and starting logging can be done using the standard procedures as described in Chapter 1

i. Installation Notes

Using a Druck UNIK 5000 Pressure Transducer, the measuring diaphragm is 34mm above the bottom of the nose cone. This means that the Zero Reference Point for a vertically installed PTI would be 34mm above the point of the cone and any offsets should take this into account.

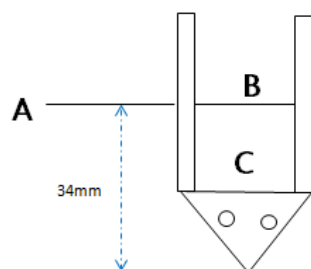
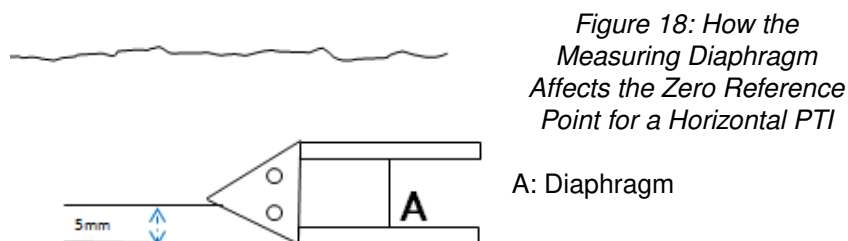


Figure 17: How the Measuring Diaphragm Affects the Druck UNIK 5000's Zero Reference Point

A: Zero Reference Point
B: Diaphragm
C: Void

The measuring diaphragm is 5mm from the edge of the transducer. This means that the

Zero Reference Point for a horizontally installed PTI would be 5mm above the edge of the transducer and any offsets should take this into account.

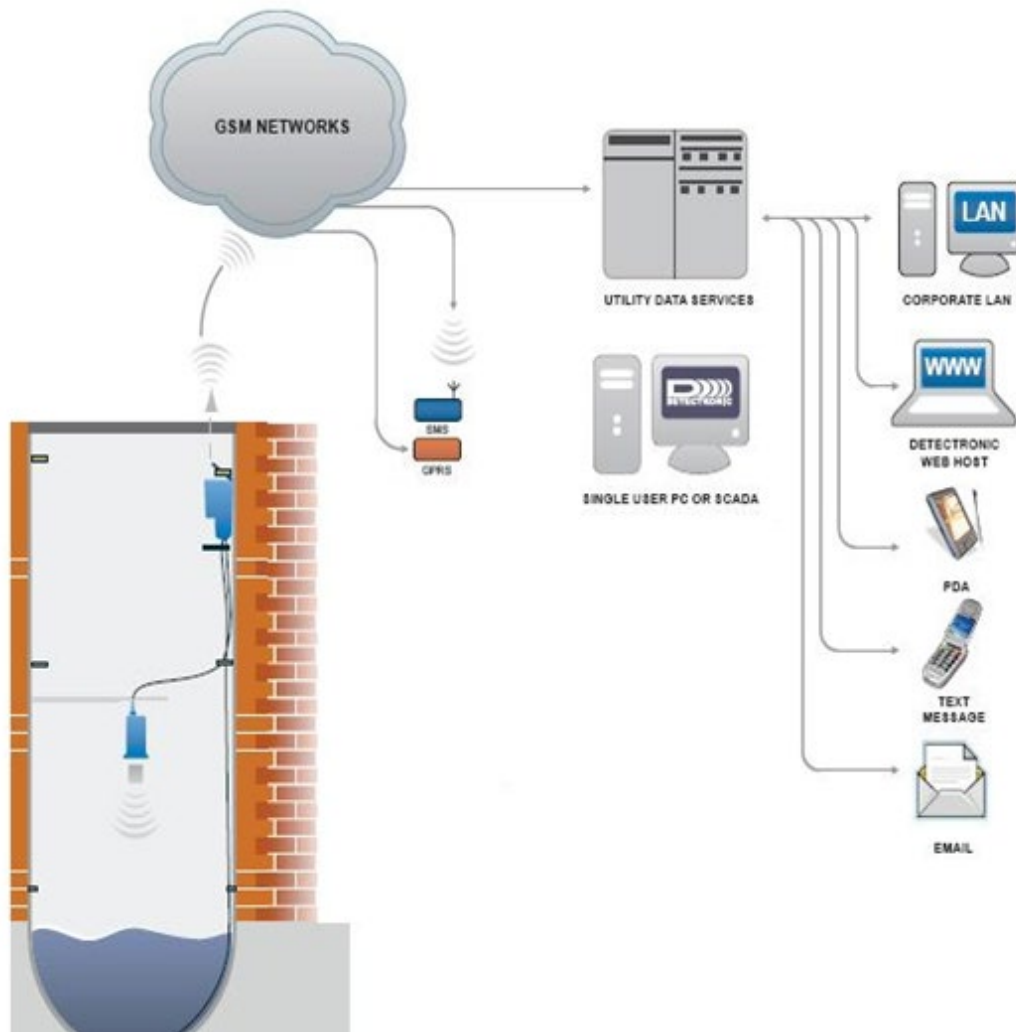


It is important when checking a pressure transducer to ensure that the void in front of or below the diaphragm is filled with water, this is particularly important for low depth measurements. Ideally the chance of an air pocket developing between the measuring diaphragm and the top of the nose cone should be eliminated. Where feasible (and specifically for installations that are not continuously immersed), a horizontal installation is preferred. For those installation where the PTI is continuously immersed, Turn the PTI within the liquid to allow the air bubble to escape.

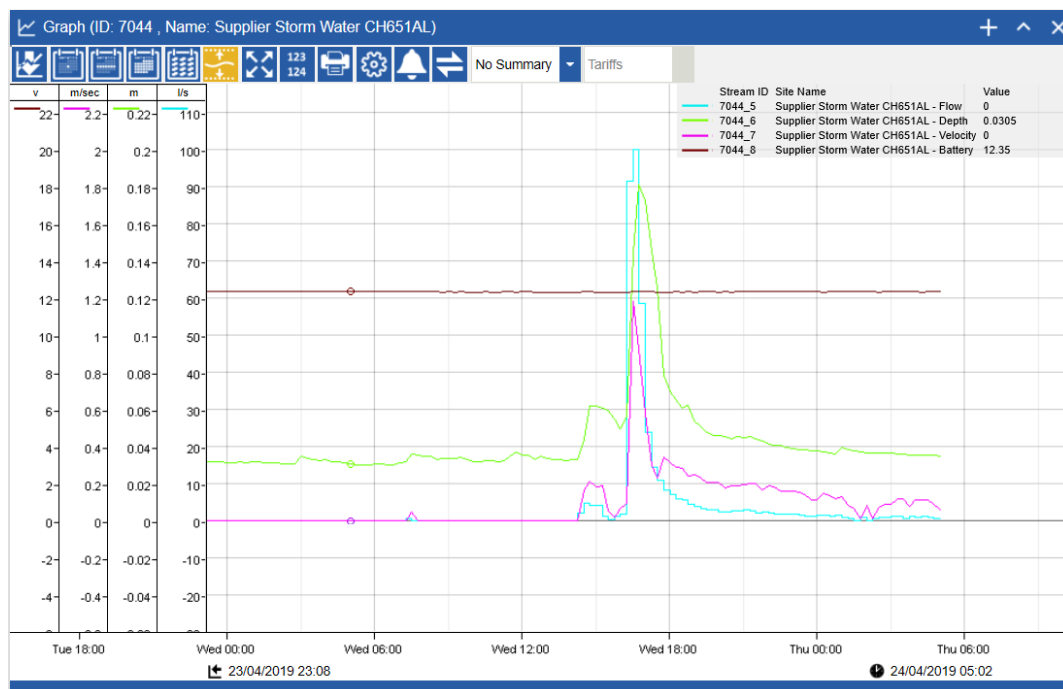
If the PTI is not already connected to a breather, ensure that the breather cover is unscrewed otherwise any readings taken during setup or testing will be erratic. The PTI cable should be supported and not allowed to simply hang under its own weight.



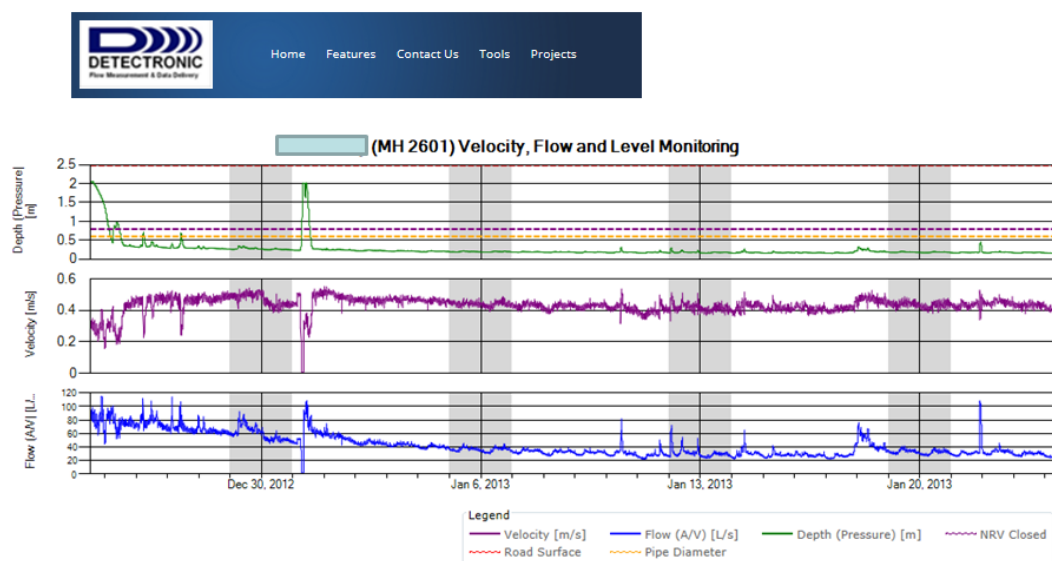
Appendix C: General Remote Comms Routing



Appendix D: Example Data



(Source: www.detecdata-en.com)



(Source: www.DetecDataPro.com)

Contact Detectronic

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For technical support and queries email: support@detectronic.org